

Juan David Muñoz-Bolaños

Rechenauerstraße 42, Innsbruck, Austria (Tirol)

+43 676 9115145

juan.munoz@i-med.ac.at

Besi Austria GmbH

Astrid Schernthaner

Innstraße 16

n.6241 Radfeld, Austria

Subject: Application for Engineering Project Manager

Dear Mrs. Schernthaner,

Driving the next wave of advanced semiconductor packaging Besi's core vision presents the kind of engineering challenges where I deal everyday. As an Optical-Physics Engineer, I am applying for the Engineering Project Manager position because I have expertise at the intersection where complex technical requirements are translated into deliverable, functional products.

A successful project manager at Besi must convert customer specifications into technical realities alongside R&D teams. During my Ph.D. at the Medical University of Innsbruck, I did not just conduct experiments; I engineered them into automated systems. I managed the complete architecture for high-precision phase retrieval, developing C++ and Python software to control adaptive hardware. Simultaneously, I coordinated the daily logistics of sourcing components from suppliers like Thorlabs and Edmund Optics to ensure our systems performed reliably.

I understand the high stakes of manufacturing timelines and quality standards. As a Junior Machine Vision Engineer at INTECOL S.A.S., I worked as part of a team implementing comprehensive machine vision solutions for industrial quality control. For example, to implement inspection systems on high-speed glass bottle lines, we combined custom illumination, optimized optics, and Halcon-based image processing algorithms to accurately verify labels, ensuring product integrity for the larger project while preventing line stoppages.

I recognize that this role requires a "one face to the customer" approach. Originally from Colombia, I have spent the last four years working on my Ph.D. in Innsbruck, fully integrating into the DACH working culture. I am fluent in English and hold a certified B2 level in German. With my Ph.D. contract concluding in May, I might be available to join your team immediately thereafter. I can seamlessly transition my current residency into a regular working permit, enabling me to drive technical negotiations and build lasting customer relationships for Besi.

I hope I can bring my hands-on problem-solving and project management skills to your team in Radfeld, and I look forward to discussing how we could build the next generation of opto-mechatronic solutions together.

Sincerely,

Juan David Muñoz-Bolaños

Optical System Engineer

Medical University of Innsbruck

Juan David Muñoz-Bolaños

Optical System Engineer

Rechenauerstraße 42, Innsbruck, Austria (Tirol)

+43 676 9115145

juan.munoz@i-med.ac.at | [LinkedIn Profile](#)



PROFESSIONAL SUMMARY

Optical System Engineer and Project Manager with a proven track record of engineering complex laboratory experiments into robust, functional products. Strong ability to bridge hardware execution with C++/Python/OpenCV control software.

TECHNICAL COMPETENCIES

Project Management & Leadership

Experienced in managing end-to-end R&D projects, translating technical requirements into integrated solutions, and leading cross-functional testing protocols.

Optical Metrology & Machine Vision

Extensive expertise in experimental optics and custom system design. Hands-on experience calibrating automated setups, interferometry, and wavefront sensing.

Software Integration & Control

Proficient in Halcon, C++, OpenCV, Python, and MATLAB for hardware interfacing, data acquisition, and real-time machine vision analysis.

PROFESSIONAL EXPERIENCE

Medical University of Innsbruck

Innsbruck, Austria

PhD Researcher - Optics & Metrology

Jan 2022 – Present

- **Product-Driven R&D:** Led the end-to-end development of advanced multiphoton microscopy setups, successfully engineering and transitioning complex laboratory experiments into a cohesive, functional product.
- **System Architecture:** Designed and integrated complete optical systems from laser sources to high-sensitivity detectors, actively managing the physical layout and optical path reliability.
- **Software & Automation (C++/Python):** Developed rigorous data acquisition and automated calibration pipelines to coordinate complex measurement protocols across motion axes and electro-optical components.
- **Hardware/Software Integration & Sourcing:** Teamed up with researchers to integrate adaptive hardware and software for high-precision phase retrieval measurements. Managed daily laboratory operations, including designing setups and sourcing optical/mechanical components from vendors like Thorlabs and Edmund Optics to guarantee reliable outcomes for peer-reviewed publications.

Leibniz Institute of Photonic Technology (IPHT)

Jena, Germany

R&D Assistant Researcher (Photonics)

Jun 2020 – Dec 2021

- **Photonics Prototyping:** Designed and constructed a dispersive 1064 nm fiber probe Raman imaging spectrometer; calibrated vibration-based point-scanning prototypes for clinical tissue applications.
- **Data Analysis & AI:** Bridged theoretical R&D with practical laboratory implementation by applying supervised machine learning algorithms to hyperspectral data for streamlined experimental characterization.

INTECOL S.A.S.

Medellín, Colombia

Junior Machine Vision System Engineer

Mar 2018 – Mar 2019

- **Project Management & Architecture:** Managed the physical design of optical inspection setups, encompassing the analysis and selection of sensors, industrial cameras, illumination schemes, and

custom optics.

- **Machine Vision Integration:** Implemented robust, industrial-grade image processing pipelines using Halcon for high-speed manufacturing environments. Successfully solved critical line-stoppage issues by engineering custom illumination systems and optimizing light intensity to reliably highlight labels on fast-moving glass bottles.

EDUCATION

Medical University of Innsbruck

Ph.D. in Adaptive Optics & Microscopy

Austria

Jan 2022 – Present

Friedrich Schiller University Jena

M.Sc. in Photonics

Germany

2019 – 2021

University of Cauca

B.Sc. in Physics Engineering

Colombia

2012 – 2018

PUBLICATIONS & AWARDS

- **Awards & Selections:** Selected for ASML and ZEISS Summer Schools; awarded COLFUTURO Grant 2026 for the best Colombian professionals.
- Muñoz-Bolaños, J. D., et al. Design of a dispersive 1064 nm fiber probe Raman imaging spectrometer and its application to human bladder resectates. *Applied Sciences*, 14(11), 4726 (2024).
- Muñoz-Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. *ACS Photonics* 12(1), 176–184 (2025).
- Sohmen, M., Muñoz-Bolaños, J. D., et al. Optofluidic adaptive optics in multi-photon microscopy. *Biomedical Optics Express* 14(4), 1562–1578 (2023).

LANGUAGES

English (Fluent / Professional),

German (B2 - Working Proficiency, incl. Mündliche Test),

Spanish (Native)

REFERENCES

Prof. Monika Ritsch-Marte

Head of Institute for Biomedical Physics

monika.ritsch-marte@i-med.ac.at

Medical University of Innsbruck

Dr. Christoph Krafft

Group leader: Spectroscopy Imaging

christoph.krafft@leibniz-ipht.de

Leibniz Institute of Photonic Technology

Passaporto № / Passport No

MUNOZ BOLANOS

COLOMBIANA

Sexo / Sex
Lugar de nacimiento / Place of birth

Fecha de expedición / Date of issue

Fecha de vencimiento / Date of expiry

G. ANTIOQUIA

Forma dei tubuli / *Peperidera* e *Aggraderia*

Dear Family

[illegible]

Zertifikat

Muñoz-Bolaños JUAN DAVID

geboren am 04.06.1994

in La Cruz / Kolumbien

hat das Modul **Schriftliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden



Datum: 16.09.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA



ÖSD ist Mitglied von



ösd

österreich schweiz deutschland

ID-Nummer: ZB2Ö25s34291

Zertifikat

Juan David MUÑOZ BALAÑOS

geboren am 04.06.1994

in La Cruz / Kolumbien

hat das Modul **Mündliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden



Datum: 11.08.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA



ÖSD ist Mitglied von



ösd

österreich schweiz deutschland



ID-Nummer: ZB2Ö25m40434



URKUNDE

Die Friedrich-Schiller-Universität Jena verleiht durch die
Physikalisch-Astronomische Fakultät
mit dieser Urkunde

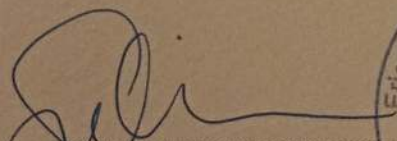
Herrn Juan David Munoz Bolanos
geb. 04.06.1994 in La Cruz (Kolumbien)
den Hochschulgrad

MASTER OF SCIENCE (M.Sc.)

Er hat die Masterprüfung (120 ECTS)
im Studiengang »Photonics«

bestanden.

Jena, 20.12.2021


Prof. Dr. Christian Spielmann
Dekan




Prof. Dr. Silvana Botti
Vorsitzende des
Prüfungsausschusses

Zertifikat

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geboren am 04.06.1994

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ÖSD ist Mitglied von



ösd

österreich schweiz deutschland



ID-Nummer: ZB2Ö25m40434

AGREEMENT FOR GUESTS

to carry out a practical training
with

Mr. **Juan David Munoz Bolanos** Date of birth: **1994-06-04**
temporarily living at **Emil-Wölk-Str. 7, 07747 Jena** (guest),

Student of the Friedrich-Schiller-University Jena, Abbe School of Photonics, gets in execution of his practical course in the research department Spectroscopy and Imaging, working group Raman and Infrared Histopathology (0104) from 2020-06-02 up to 2020-12-31 admission to the necessary rooms within Leibniz-IPHT.

The use of further devices, media and other infrastructure takes place in agreement with the Heads of the respective departments.

Mr. Juan David Munoz Bolanos has to observe the security regulations for working and other regulations valid in Institute as well as instructions of the laboratory and clean room responsible persons of the Leibniz-IPHT.

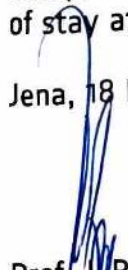
Mr. Juan David Munoz Bolanos has to treat all affairs and procedures strictly confidential. Copies, duplicates and others of business and technical information of each kind are forbidden, unless these are sanctioned by the responsible Department Heads.

Publications, reports and lectures about topics which belong together with the fields of work of Leibniz-IPHT require the previous written agreement of the Executive Board.

The liability of Mr. Juan David Munoz Bolanos is limited to cases of wilful action and gross negligence.

Mr. Juan David Munoz Bolaos is insured against accidents on the Friedrich-Schiller-University Jena, Abbe School of Photonics in the context of his/her practical course during his/her length of stay at Leibniz-IPHT.

Jena, 18 May 2020


Prof. Dr. Popp
Chairman
of Leibniz-IPHT


F. Sonderrmann
Executive Member
of Leibniz-IPHT


Juan David Munoz Bolanos
Guest

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Jena, 22 September 2021

Prof. J. Popp
Chairman
of Leibniz-IPHT

F. Sondermann
Executive Member
of Leibniz-IPHT

Juan David Munoz Bolanos
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AGREEMENT FOR GUESTS to carry out a practical training with

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Student of the Friedrich-Schiller-University Jena, Abbe School of Photonics, gets in execution of his practical course in the research department Spectroscopy and Imaging, working group Raman and Infrared Histopathology (0104) from 2021-01-01 up to 2021-09-30 admission to the necessary rooms within Leibniz-IPHT.

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Jena, 13 November 2020


Prof. J. Popp
Chairman
of Leibniz-IPHT


F. Sondermann
Executive Member
of Leibniz-IPHT


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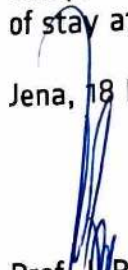
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Jena, 18 May 2020


Prof. Dr. Popp
Chairman
of Leibniz-IPHT


F. Sonderrmann
Executive Member
of Leibniz-IPHT


Juan David Munoz Bolanos
Guest

Juan Munoz Bolanos

has participated in the

**ZEISS European
Summer School**

Lithography Optics &
Semiconductor Technologies

26 - 27 June 2025

ZEISS Semiconductor Manufacturing Technology, Oberkochen, Germany

Thomas Marschner

Dr. Thomas Marschner
Program Manager Funding Projects

Thomas Stammer

Dr. Thomas Stammer
CTO & Head of Product Strategy



INTECOL

OPTIMIZING THE RHYTHM OF THE WORLD

Medellin, February 4, 2019

TO WHOM IT MAY CONCERN

Integración Tecnológica Industrial Intecol S.A.S. with Tax ID (NIT) 811.041.410-4 certifies that Mr. JUAN DAVID MUÑOZ BOLAÑOS, identified with Citizenship ID No. 1.061.772.278, has been employed by this company since July 3, 2018, under an indefinite term contract, serving in the position of Junior Project Engineer;

earning a salary of One million five hundred thousand pesos (\$1,500,000) plus three hundred thousand pesos (\$300,000) mobility allowance.

I will gladly address any additional information at the phone number 316 3322.

We appreciate your attention,

INTECOL

NIT. 811.041.410-4 - Tel.: 3163822

LENY ADRIANA RUIZ PINO

Administrative and Financial Director

Integración Tecnológica Industrial Intecol S.A.S.
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Medellin Calle 42 #63c-82
PBX. (574) 316 3322 Cell: (57) 301 430 2532